# PolicyCut

Write your privacy policy in plain English — review, redact and audit your video, fully local on AMD Radeon.

**AMD Radeon Hackathon 2026**

Track 1 · Multimodal Content Creation Tools

## The problem

Real footage is full of personal data: bystander faces, plates, ID cards, screens, spoken phone numbers and addresses. Manual redaction takes hours; cloud services require uploading exactly the footage you must protect; single-model demos cover one modality with no review and no audit.

## One sentence in, evidence out

```
"Pixelate every face, license plate,
 identity document and screen.
 Beep phone numbers, email addresses
 and street addresses."
```

Upload video (≤60 s)

▼

Interpret policy → structured rules you confirm

▼

Automatic analysis (vision + audio, on Radeon)

▼

**HUMAN REVIEW — confirm / fix / add on a dual-track timeline**

▼

Render → redacted MP4 + redacted SRT + audit.json

## All local, all on Radeon

**Policy → JSON rules**
Qwen3-4B-Instruct, strict schema, deterministic keyword fallback

**Open-vocab detection**
Grounding DINO tiny (Transformers port — no CUDA custom ops)

**Small-face assist**
YuNet (OpenCV Zoo, CPU assist)

**Mask tracking**
SAM 2.1 hiera-small, video propagation, SAM2_BUILD_CUDA=0

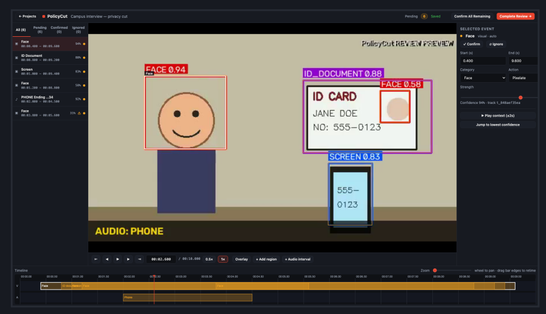
**Speech-to-text**
Whisper large-v3-turbo, word-level timestamps

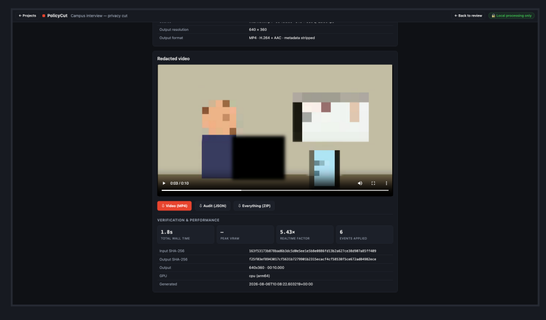
**Spoken-PII locate**
regex + spoken-form normalization ("five five five … at example dot com")

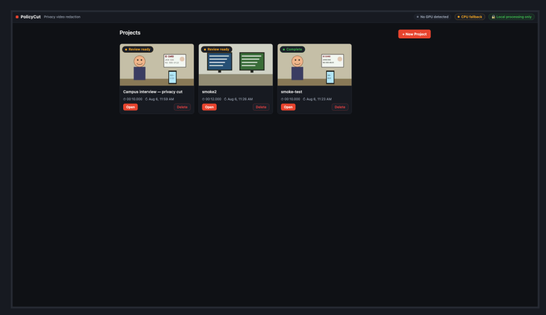
## Fail toward privacy

- Empty SAM mask → padded box, never a hole
- Mask dilate 5–12 px + feather + safety box
- No word timestamps → whole segment muted
- Audio spans padded –150/+250 ms, gaps merged
- Unknown policy terms surfaced, never guessed
- Export blocked until every event reviewed

## A real editor, not a demo page






## Every export ships evidence

```
redacted.mp4  H.264/AAC, metadata stripped
redacted.srt  [REDACTED: PHONE|EMAIL|ADDRESS]
audit.json    models, precision, timings, VRAM,
              every event + human edits,
              zero PII plaintext inside
SHA-256 of input and output, ZIP bundle
```

## Measured on Radeon PRO W7900

GPU: AMD Radeon PRO W7900 (48 GB)
ROCm/HIP: ROCm PyTorch build
PyTorch: platform ROCm wheels
Precision: FP16 (per-stage FP32 fallback)

**Measured benchmark tables:**

- Grounding DINO keyframes/s
- SAM 2.1 propagation frames/s
- Whisper real-time factor
- End-to-end real-time factor
- Peak VRAM per stage

→ `artifacts/benchmarks/*.json`
→ `docs/benchmark.md`

## Honest Radeon engineering

- --require-rocm: core stages refuse silent
- CPU fallback — a Radeon run is provable
- Sequential model load/unload per stage
- torch.cuda.synchronize() around all timings
- Peak VRAM recorded per stage in audit.json
- ROCm-clean models: no CUDA custom kernels
- Zero external API calls during processing

**PolicyCut is a human-in-the-loop privacy editor with an auditable, fully local workflow — not a guarantee of legal compliance.**